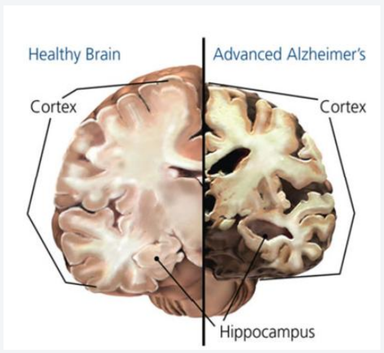
Using Artificial Intelligence and Deep Neural Machine Learning Networks to Analyse High-Dimensional Datasets Related to Biological Processes in Infection and Inflammation

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LAB POSTERS



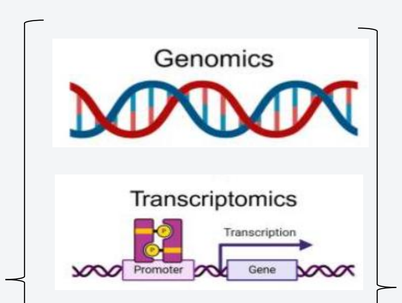
1 Background

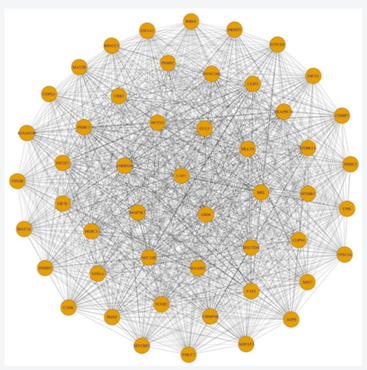
Full-scale analysis of an Alzheimer's disease (AD) brain is a daunting task, as the pathology involves stage-dependent, multi-pathway dysregulation that single-marker methods currently capture poorly.

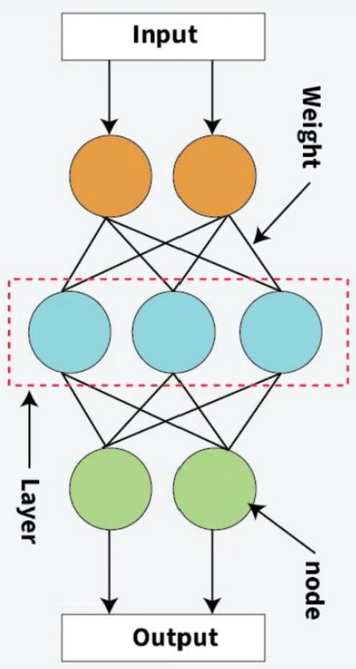
We propose a more **network-centric AI framework** to learn multivariate signatures of AD progression and convert them into interaction network graphs that are patient-personalised. This work utilises the highly diverse **ROSMAP** dataset, which contains clinical and epidemiological longitudinal data, multi-omics data, and accompanying patient lifestyle data points.

Progression of tau (green) pathology across Braak stages (I–VI) in Alzheimer's disease











Extract information to create biological understanding

2 Key Approaches

2.1 Data Integration
Synolytic networks combine multiple biological measurements (e.g., proteins, genes, or clinical variables) into a single network, letting features be examined together rather than independently.

2.2 Network Construction
Features with similar behaviour across samples are connected, forming a network where nodes represent biological variables and edges represent statistically significant associations.

2.3 Pattern Identification
The network is analysed to identify important modules, highly connected nodes (hubs), and key interactions that may distinguish biological groups or disease states.

2.4 Biological Interpretation
The identified network structures are used to generate biological insights, helping to uncover potential disease behaviour, pathways, and mechanisms that can support further experimental validation and clinical research. These methods can then be applied to other disease datasets (Cancer, Virology, Genomics).

3 Method and Results

3.1 **ROSMAP** dataset:
(The **R**eligious **O**rders **S**tudy and **M**emory and **A**ging **P**roject)
- **400 patients**, with **8,817 proteins** per patient.
- Raw dataset contains **257,587 data gaps**.
- **Expected analysis** = 38,865,936 rows or **389 CSV files**.

3.2 **Imputation type: MissForest**
Missingness type: Random.
Description: Random forest-based iterative imputation that predicts missing values using relationships between variables.
Pros: High accuracy, handles complex/non-linear data, suitable for mixed data types.
Cons: Computationally intensive.

3.3 **Imputation type: KNN**
Missingness type: Random.
Description: Estimates missing values using the values of the most similar samples (nearest neighbours).
Pros: Simple, preserves local data structure, no distribution assumptions.
Cons: Sensitive to noisy data.

3.4 **Imputation type: QRILC / MinProb**
Missingness type: Non-random.
Description:
QRILC: Uses quantile regression to impute values below detection limit.
MinProb: Replaces missing values with small random and low-intensity distribution.
Pros: Designed for proteomics low-abundance data, preserves distribution.
Cons: Can bias low-end distribution.

4 Extracting Biological Meaning

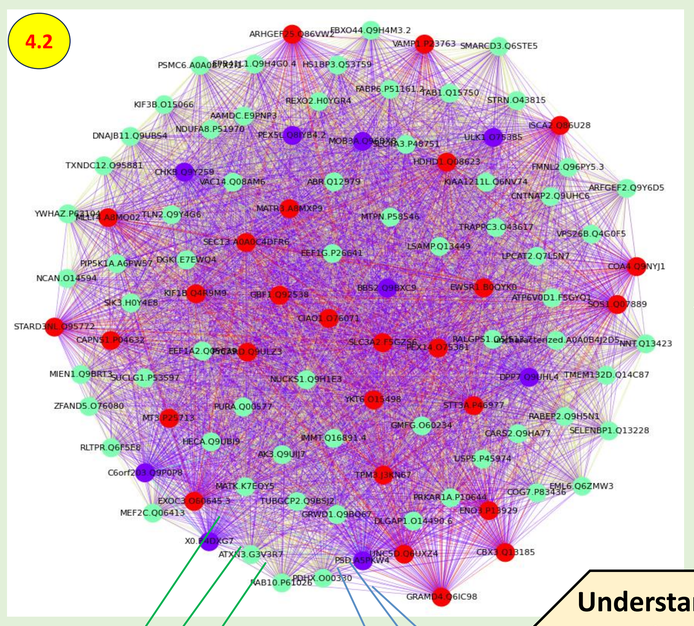
Patient-personalised protein networks
- **Focus**: Multilayered Alzheimer's Disease dataset (**ROSMAP**).
- **Tools**: R/Python code, High-performance computing, Synolytic protein networks.
- **Goal**: Identify hubs, clusters, and extract biologically meaningful information.
- **Generate new insights**: **Disease-stage** (Braak stage, CERAD score, and cognitive decline trajectories), **Benchmarking** (compare network approaches against univariate methods), **Pathway analysis** (identify biological pathways such as synaptic and immune processes).

4.1

Patient A	1.12
Patient B	0.41
Patient C	2.03
Patient D	0.77
Patient E	1.58

Disconnected tables

4.2



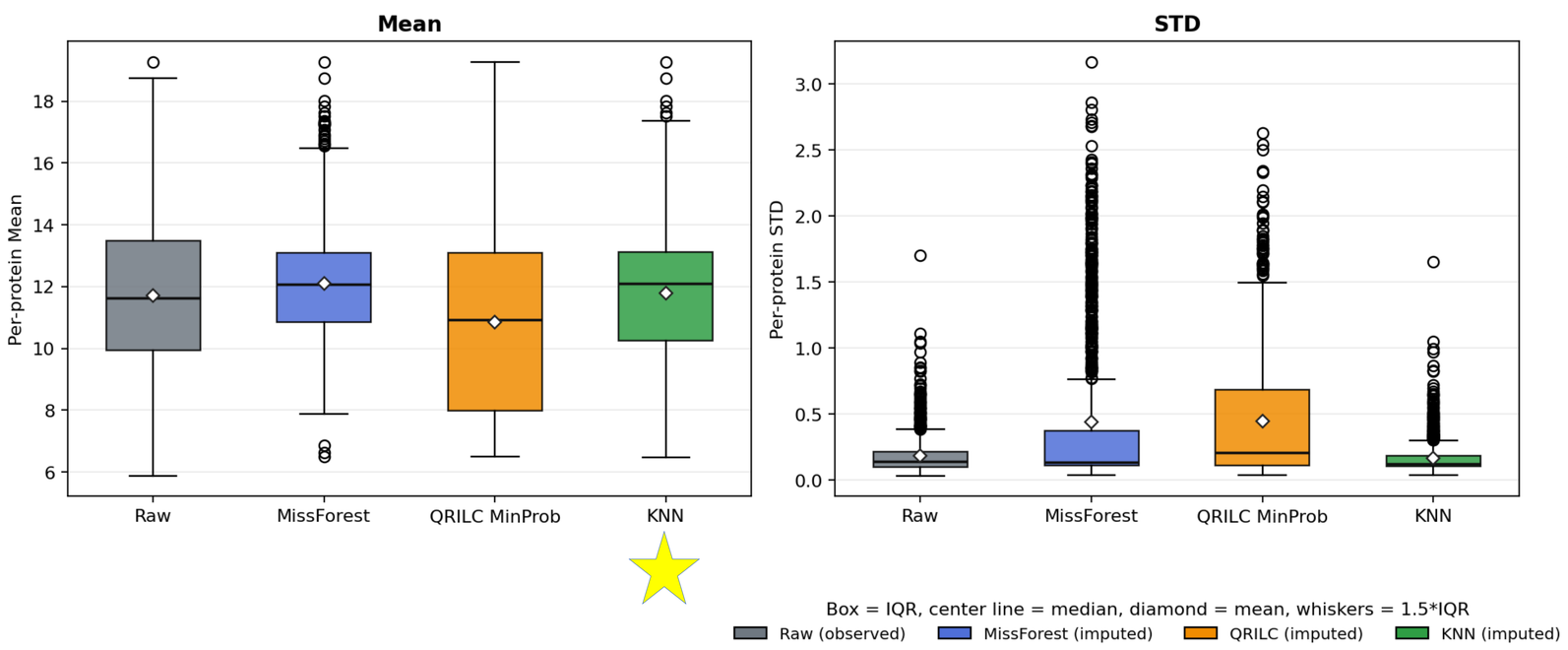
Connected network

Understand molecular and system-level disease changes

4.3

APOE: Key in amyloid processing.
APP: Amyloid precursor protein.
Tau: Core pathology in neurofibrillary tangles.
PSEN1: Amyloid-beta generation pathway.

Imputation results: Per-protein Mean and STD, on log2 scale

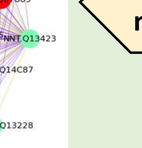


Box plots compare per-protein Mean and STD, on log2 scale for the imputation methods: MissForest, QRILC, and KNN versus raw dataset.

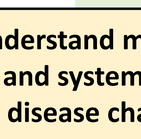
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Alzheimer's Disease • Multi-Omics • Network Analysis • Interpretable AI • ROSMAP • Personalised Networks • Extracting Biological Meaning